

Mini project

GitOps:

Continuous integration with GitHub Actions & Continuous Delivery with ArgoCD

- This project contains a hands-on implementation of a robust GitOps pipeline to automate the deployment and scaling of a simple application. By separating application code from infrastructure configuration, the project demonstrates real-world cloud-native production practices including automated CI/CD synchronization, infrastructure-as-code configuration management, and zero-downtime progressive delivery.
- Project Structure:
 - GitOps Repository Segmentation & Structure
 - Continuous Integration via GitHub Actions (Docker Buildx & Dynamic Tagging)
 - Automated Manifest Updates & Git Repository Synchronization
 - ArgoCD Declarative Application Deployment & State Reconciliation
 - Progressive Delivery Implementation (Canary/Blue-Green Deployments via Argo Rollouts)

- Repositories :

app-repo : <https://github.com/MaZeNNaSSer22/Ci-Using-GithubActions.git>

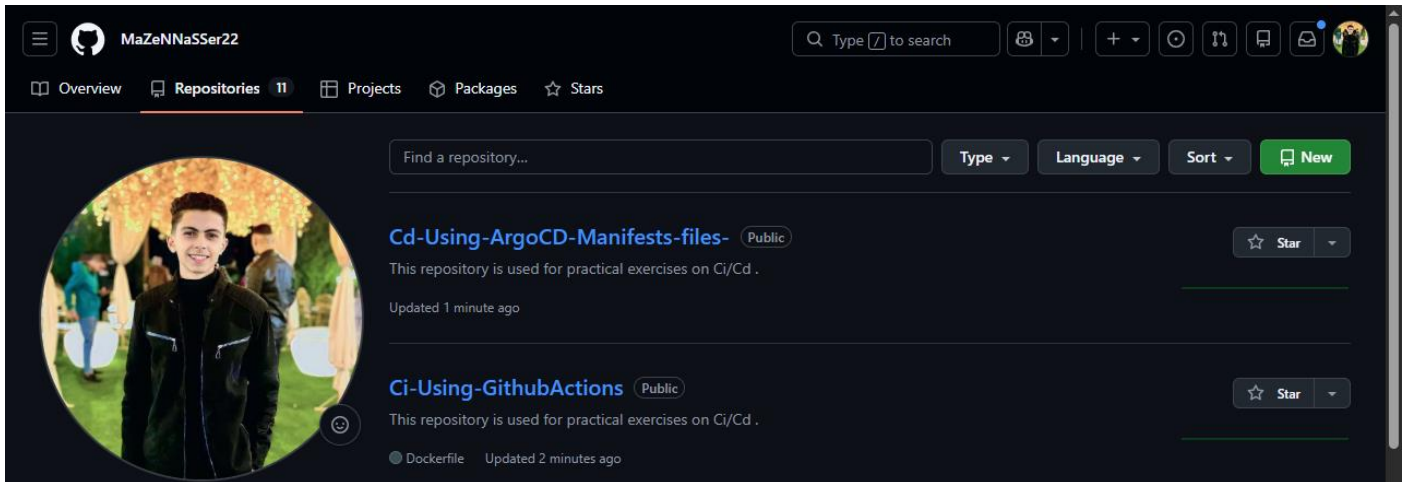
manifests-repo : <https://github.com/MaZeNNaSSer22/Cd-Using-ArgoCD-Manifests-files-.git>

Docker-Hub : <https://hub.docker.com/repository/docker/mazennasser/http-app/general>

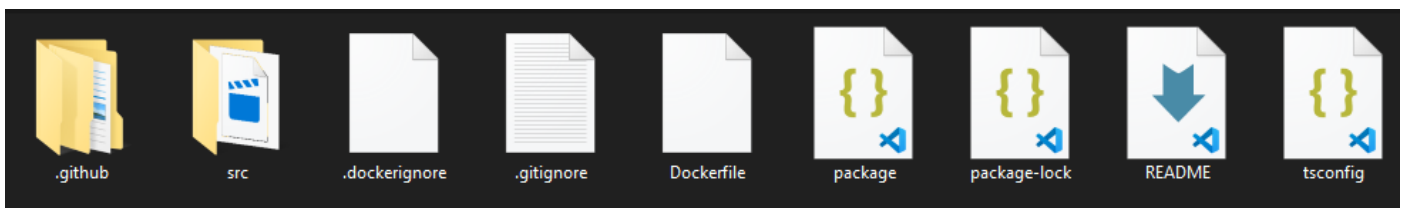
1- Prepare the cluster

```
mazen22@MN22-VM:~$ minikube start
minikube v1.38.0 on ubuntu 24.04
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.49 ...
* Restarting existing docker container for "minikube" ...
* minikube 1.38.1 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.38.1
* To disable this notice, run: 'minikube config set WantUpdateNotification false'
```

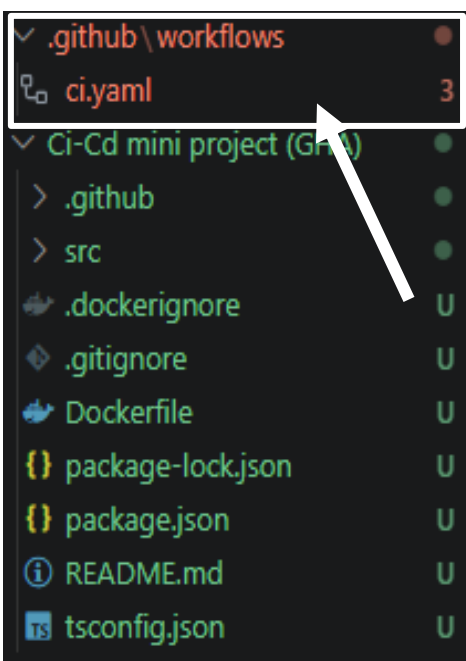
2- Create Two repositories (app-repo & manifest-repo)



3- Prepare the App with its dependencies & Dockerfile



4- Write the Ci workflow



Workflow steps:

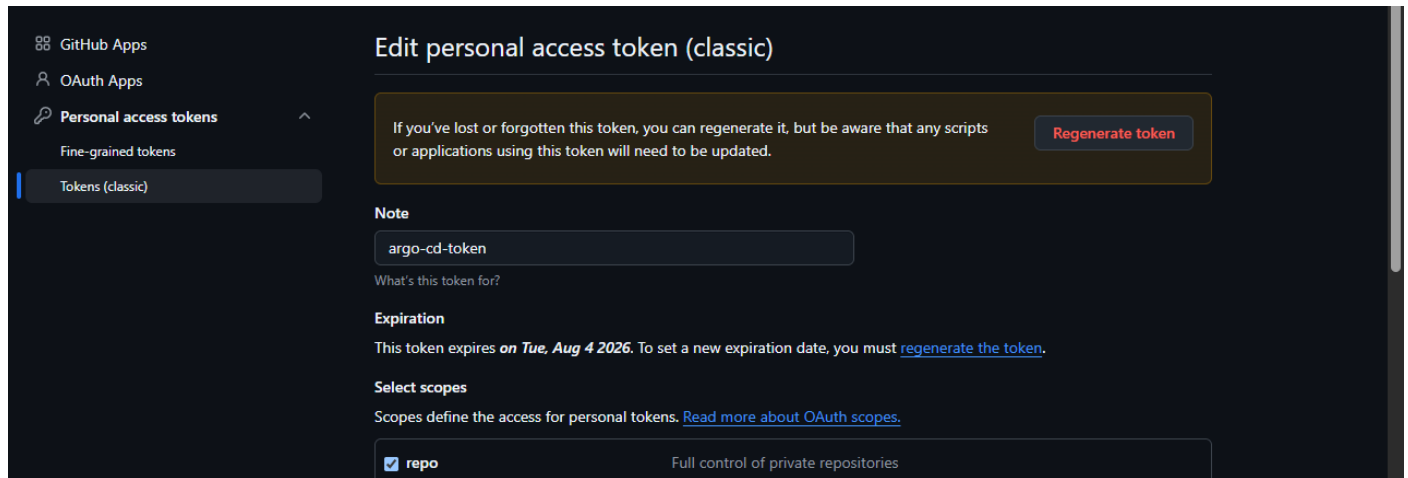
- 1- Test & build
- 2- Login to Docker hub
- 3- Push the image to The Hub
- 4- Clone manifest-repo
- 5- Update Image Tag
- 6- Create Pull Request

This need to add some secrets such as

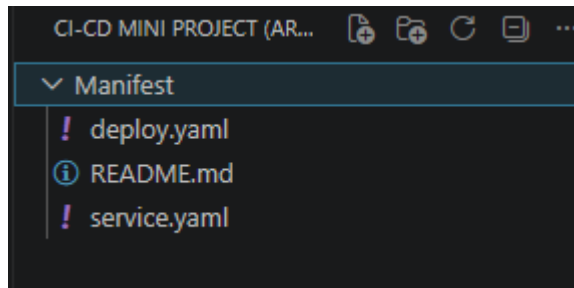
- 1- DOCKER_USERNAME
- 2- DOCKER_PASSWORD
- 3- GITOPS_PAT

This **GITOPS_PAT** is personal access token (classic) : to allow the app-repo to clone and raise a pull-request on manifest-repo

(settings → Developer settings → personal access token → Token classic)



5- Write The manifests files



6- Install ArgoCD in the cluster

```
mazen22@MN22-VM:~$ kubectl create namespace argocd
kubectl apply -n argocd --server-side --force-conflicts -f https://raw.githubusercontent.com/argoproj/argo-cd/v3.4.4/manifests/install.yaml
namespace/argocd created
customresourcedefinition.apiextensions.k8s.io/applications.argoproj.io serverside-applied
customresourcedefinition.apiextensions.k8s.io/applicationsets.argoproj.io serverside-applied
customresourcedefinition.apiextensions.k8s.io/appprojects.argoproj.io serverside-applied
serviceaccount/argocd-application-controller serverside-applied
serviceaccount/argocd-applicationset-controller serverside-applied
serviceaccount/argocd-dex-server serverside-applied
serviceaccount/argocd-notifications-controller serverside-applied
serviceaccount/argocd-redis serverside-applied
serviceaccount/argocd-repo-server serverside-applied
serviceaccount/argocd-server serverside-applied
role.rbac.authorization.k8s.io/argocd-application-controller serverside-applied
role.rbac.authorization.k8s.io/argocd-applicationset-controller serverside-applied
role.rbac.authorization.k8s.io/argocd-dex-server serverside-applied
role.rbac.authorization.k8s.io/argocd-notifications-controller serverside-applied
role.rbac.authorization.k8s.io/argocd-redis serverside-applied
role.rbac.authorization.k8s.io/argocd-server serverside-applied
clusterrole.rbac.authorization.k8s.io/argocd-application-controller serverside-applied
clusterrole.rbac.authorization.k8s.io/argocd-applicationset-controller serverside-applied
clusterrole.rbac.authorization.k8s.io/argocd-server serverside-applied
rolebinding.rbac.authorization.k8s.io/argocd-application-controller serverside-applied
rolebinding.rbac.authorization.k8s.io/argocd-applicationset-controller serverside-applied
```

7- Edit argocd-server Service to be NodePort

```
mazen22@MN22-VM:~$ kubectl -n argocd edit svc argocd-server
service/argocd-server edited
```

```
sessionAffinity: None
type: NodePort
```

8- Sign in to ArgoCD UI

```
mazen22@MN22-VM:~$ kubectl -n argocd get svc argocd-server
NAME      TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
argocd-server  NodePort    10.108.97.39   <none>         80:30769/TCP,443:32440/TCP  70s
```

192.168.49.2

```
mazen22@MN22-VM:~$ minikube ip
192.168.49.2
```

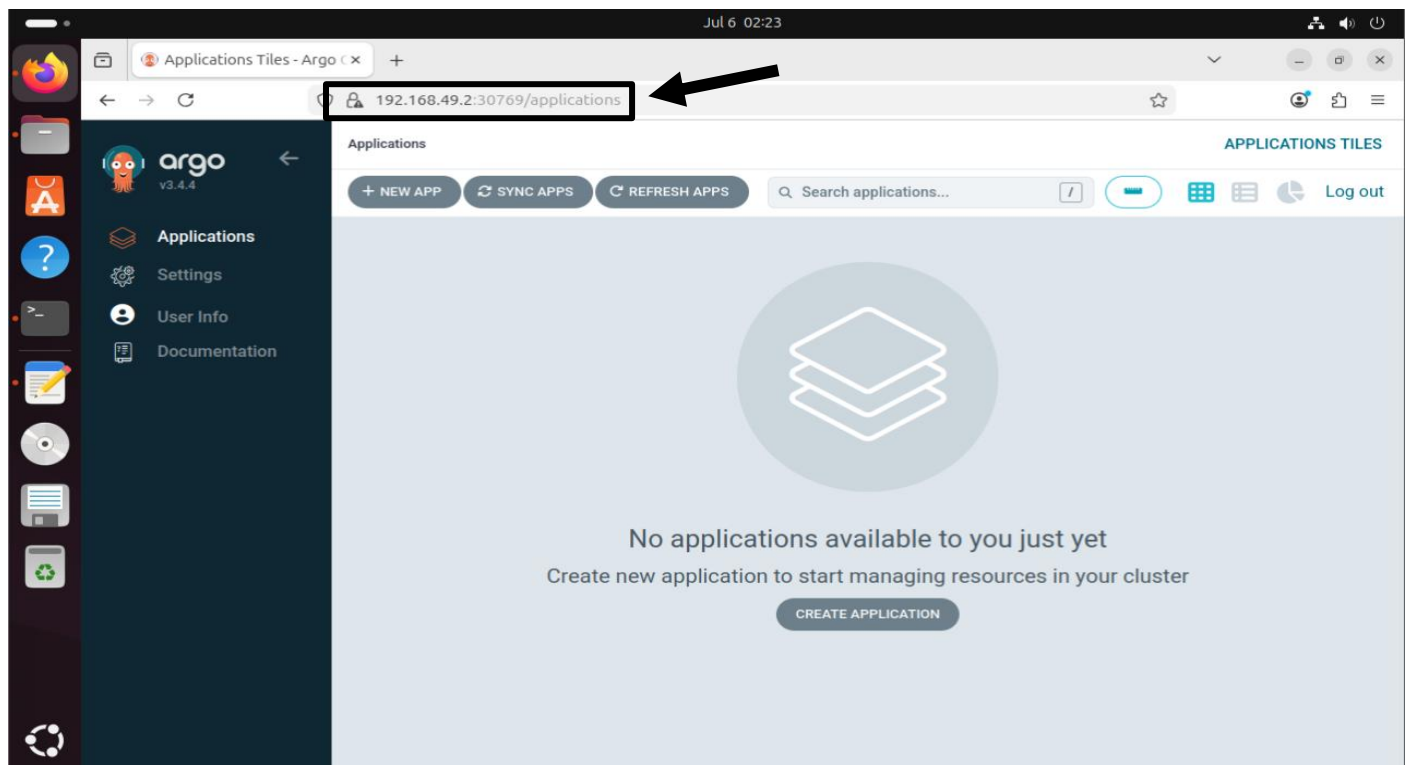
```
mazen22@MN22-VM:~$ kubectl -n argocd get secrets argocd-initial-admin-secret -o json | jq .data.password -r | base64 -d
```

Annotations:

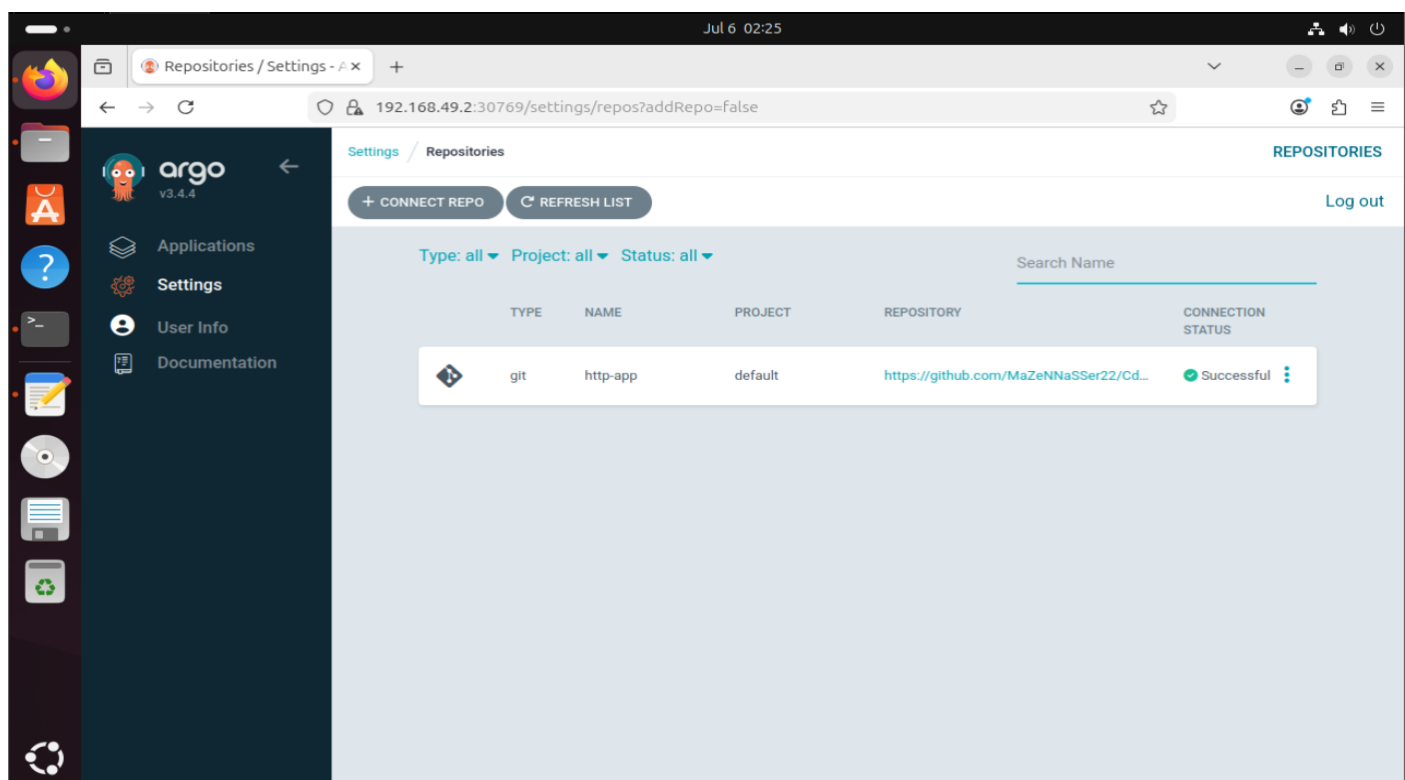
- Arrow pointing to `argocd-server` in the first command: `argocd-server`
- Arrow pointing to `80:30769/TCP` in the first command: `80:30769/TCP`
- Arrow pointing to the output of the second command: `192.168.49.2`
- Arrow pointing to the output of the third command: `192.168.49.2`
- Arrow pointing to the output of the fourth command: `192.168.49.2`

Password of Argo account

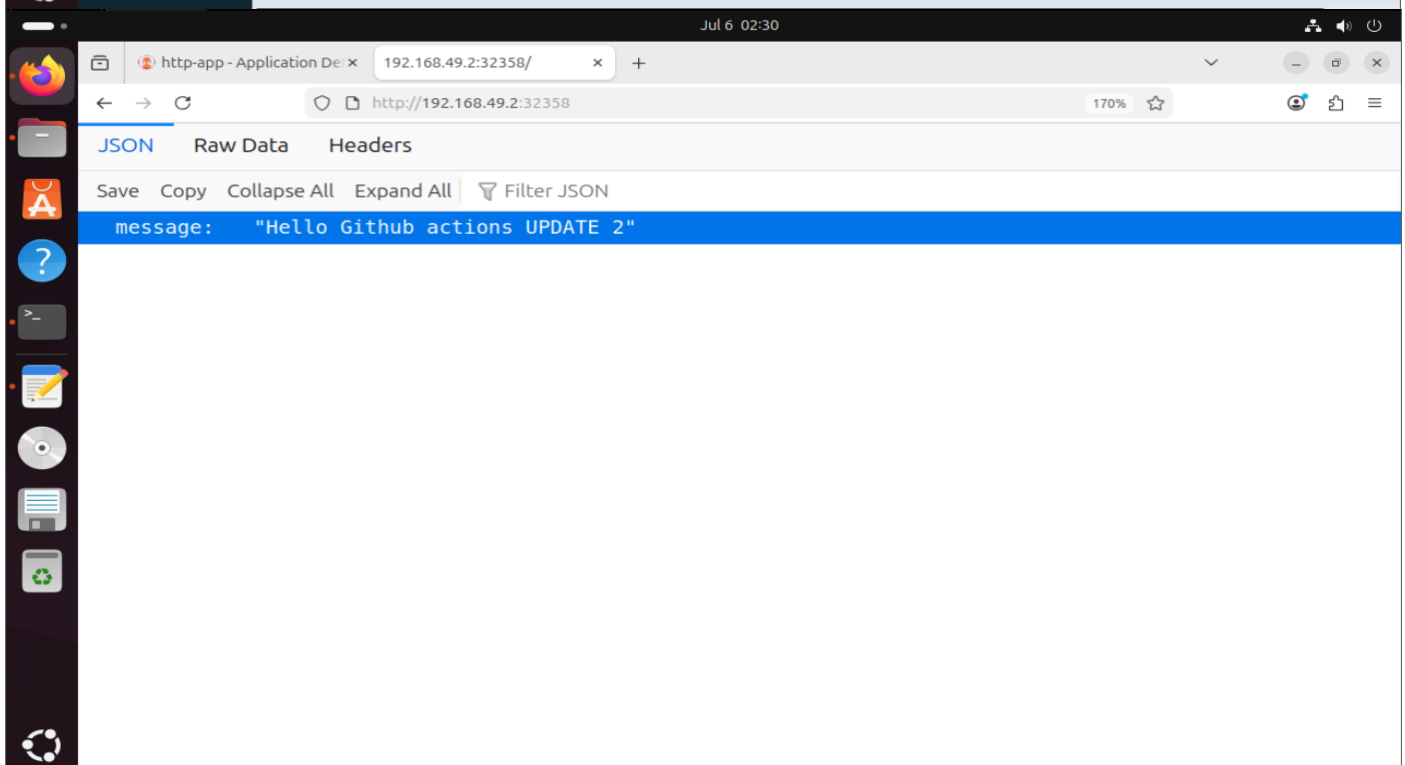
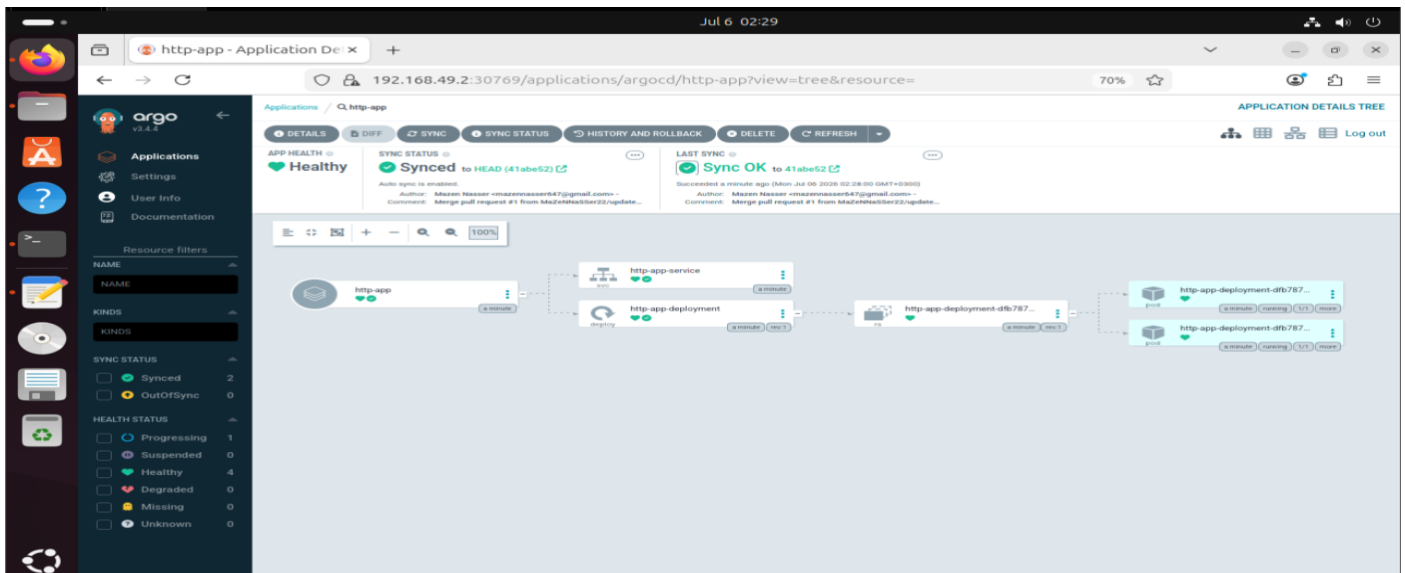
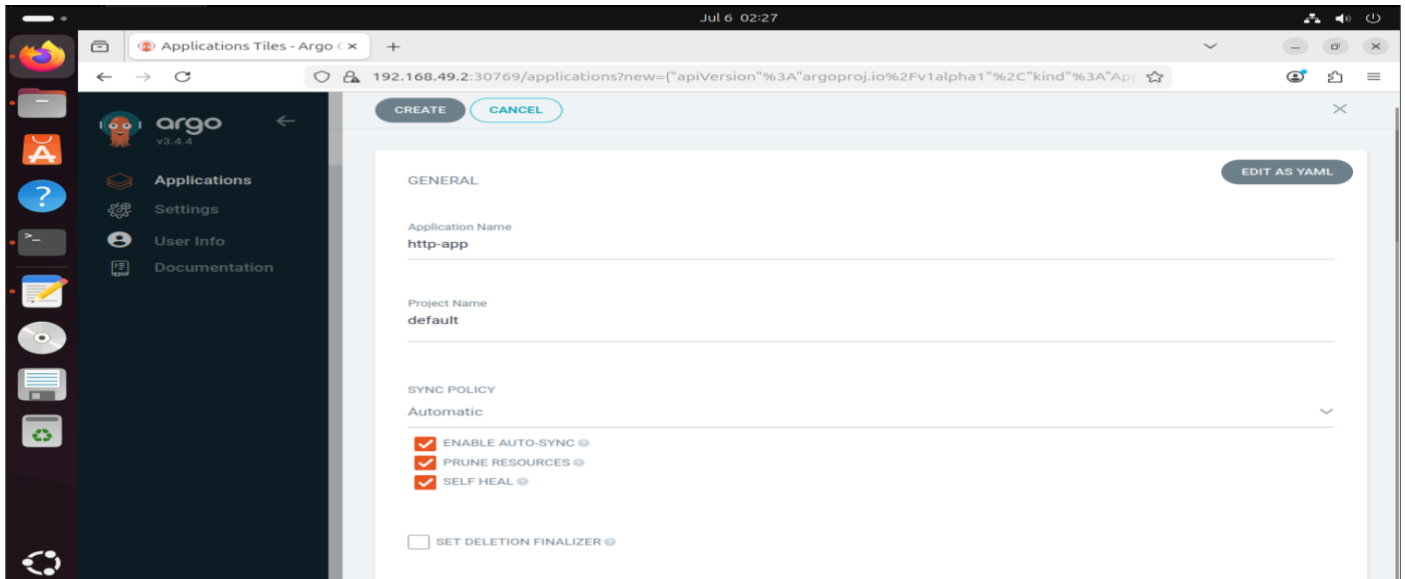
To access the ArgoCD UI → <http://192.168.49.2:30769>



9- Add the manifest-repo to ArgoCD repositories



10- Create an application with these manifests



11- Try To Check the functionality of Ci/Cd workflow

```
src > TS app.ts > ...
...
1  import http from "node:http";
2
3  const PORT = 3000;
4
5  function bootstrap() {
6    const server = http.createServer((req, res) => {
7      const { method, url } = req;
8
9      if (method === "GET" && url === "/") {
10        res.writeHead(200, { "Content-Type": "application/json" });
11        res.end(JSON.stringify({ message: "Hello Github actions UPDATE 2" }));
12        return;
13      }
14
15      if (method === "GET" && url === "/message2") {
16        res.writeHead(200, { "Content-Type": "application/json" });
17        res.end(JSON.stringify({ message: "new message" }));
18        return;
19      }
20
21      res.writeHead(404, { "Content-Type": "application/json" });
22      res.end(JSON.stringify({ error: "Not Found" }));
23    });
24
25    server.listen(PORT, () => {
26      console.log(`Server running at http://localhost:${PORT}`);
27    });
28  }
29
30  bootstrap();
  You, 5 hours ago • first try

src > TS app.ts > ...
You, 5 hours ago | 1 author (You)
1  import http from "node:http";
2
3  const PORT = 3000;
4
5  function bootstrap() {
6    const server = http.createServer((req, res) => {
7      const { method, url } = req;
8
9      if (method === "GET" && url === "/") {
10        res.writeHead(200, { "Content-Type": "application/json" });
11        res.end(
12          JSON.stringify({ message: "Hello Github actions UPDATE 3 (By Mazen)" }),
13        );
14        return;
15      }
16
17      if (method === "GET" && url === "/message2") {
18        res.writeHead(200, { "Content-Type": "application/json" });
19        res.end(JSON.stringify({ message: "new message" }));
20        return;
21      }
22
23      res.writeHead(404, { "Content-Type": "application/json" });
24      res.end(JSON.stringify({ error: "Not Found" }));
25    });
26
27    server.listen(PORT, () => {
28      console.log(`Server running at http://localhost:${PORT}`);
29    });
30  }
31
32  bootstrap();
33
  You, 59 seconds ago • Uncommitted changes
```

⌵

MaZeNNaSSer22 / Ci-Using-GithubActions

🔍 Type / to search

🔧

+

🕒

🔄

📄

📧

👤

<> Code ⌚ Issues 🔄 Pull requests 🏠 Agents ⚙️ Actions 📁 Projects 📖 Wiki 🛡️ Security and quality 📈 Insights ⚙️ Settings

← HTTP APP workflow

🔴 Change the http page content (check Cd functionality) #13 Cancel workflow ⋮

🏠 Summary

All jobs

🔴 test_and_build

Run details

🕒 Usage

📄 Workflow file

Triggered via push now

MaZeNNaSSer22 pushed -> 7a2066e main

Status

In progress

Total duration

—

Artifacts

—

ci.yaml

on: push

🔴 test_and_build 18s

🔄 - +

⌵

MaZeNNaSSer22 / Ci-Using-GithubActions

🔍 Type / to search

🔧

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← HTTP APP workflow

🟢 Change the http page content (check Cd functionality) #13 Re-run all jobs ⋮

🏠 Summary

All jobs

🟢 test_and_build

Run details

🕒 Usage

📄 Workflow file

Triggered via push now

MaZeNNaSSer22 pushed -> 7a2066e main

Status

Success

Total duration

32s

Artifacts

—

ci.yaml

on: push

🟢 test_and_build 29s

🔄 - +

MaZeNNaSSer22 / Cd-Using-ArgoCD-Manifests-files-

Q Type to search

+

<> Code Issues Pull requests 1 Agents Actions Projects Wiki Security and quality Insights Settings

Label issues and pull requests for new contributors
Now, GitHub will help potential first-time contributors [discover issues](#) labeled with [good first issue](#)

Dismiss

Filters Q is:pr is:open

Labels 9

Milestones 0

New pull request

1 Open 1 Closed

Author Label Projects Milestones Reviews Assignee Sort

Deploy: Update http-app image to build 13

#2 opened now by MaZeNNaSSer22 (Owner)

MaZeNNaSSer22 / Cd-Using-ArgoCD-Manifests-files-

Q Type to search

+

<> Code Issues Pull requests 1 Agents Actions Projects Wiki Security and quality Insights Settings

Deploy: Update http-app image to build 13 #2

Ready to merge Code

Open

MaZeNNaSSer22 wants to merge 1 commit into [main](#) from [update-image-13](#)

Conversation 0

Commits 1

Checks 0

Files changed 1

+1 -1

MaZeNNaSSer22 commented now

Owner

This PR was automatically generated by GitHub Actions to update the application image version to 13.

chore: update http-app image tag to build 13

d04a38e

No conflicts with base branch

Merging can be performed automatically.

Merge pull request You can also merge this with the command line. [View command line instructions.](#)

Reviewers

No reviews

Still in progress? [Convert to draft](#)

Assignees

No one—[assign yourself](#)

Labels

None yet

Projects

None yet

Milestone

No milestone

Jul 6 02:33

http-app - Application De x 192.168.49.2:32358/ x +

192.168.49.2:30769/applications/argocd/http-app?view=tree&resource=

argocd v3.4.4

Applications

Settings

User Info

Documentation

Resource filters

NAME

KINDS

SYNC STATUS

HEALTH STATUS

APPLICATION DETAILS TREE

DETAILS DIFF SYNC SYNC STATUS HISTORY AND ROLLBACK DELETE REFRESH

APP HEALTH **Healthy**

SYNC STATUS **OutOfSync** from HEAD (b07066a)

LAST SYNC **Syncing**

Auto sync is enabled.

Author: Mazen Nasser <mazenasser647@gmail.co...>

Comment: Merge pull request #2 from MaZeNnASer2...

Running a few seconds ago (Mon Jul 06 2026 02:33:57 GMT+0300)

waiting to start

http-app

http-app-service

http-app-deployment

http-app-deployment

Jul 6 02:34

http-app - Application De x 192.168.49.2:32358/ x +

192.168.49.2:30769/applications/argocd/http-app?view=tree&resource= 70%

argocd v3.4.4

Applications

Settings

User Info

Documentation

Resource filters

NAME

KINDS

SYNC STATUS

HEALTH STATUS

APPLICATION DETAILS TREE

DETAILS DIFF SYNC SYNC STATUS HISTORY AND ROLLBACK DELETE REFRESH

APP HEALTH **Healthy**

SYNC STATUS **Synced** to HEAD (b07066a)

LAST SYNC **Sync OK** to b07066a

Auto sync is enabled.

Author: Mazen Nasser <mazenasser647@gmail.co...>

Comment: Merge pull request #2 from MaZeNnASer22/update...

Succeeded a few seconds ago (Mon Jul 06 2026 02:33:58 GMT+0300)

Author: Mazen Nasser <mazenasser647@gmail.co...>

Comment: Merge pull request #2 from MaZeNnASer22/update...

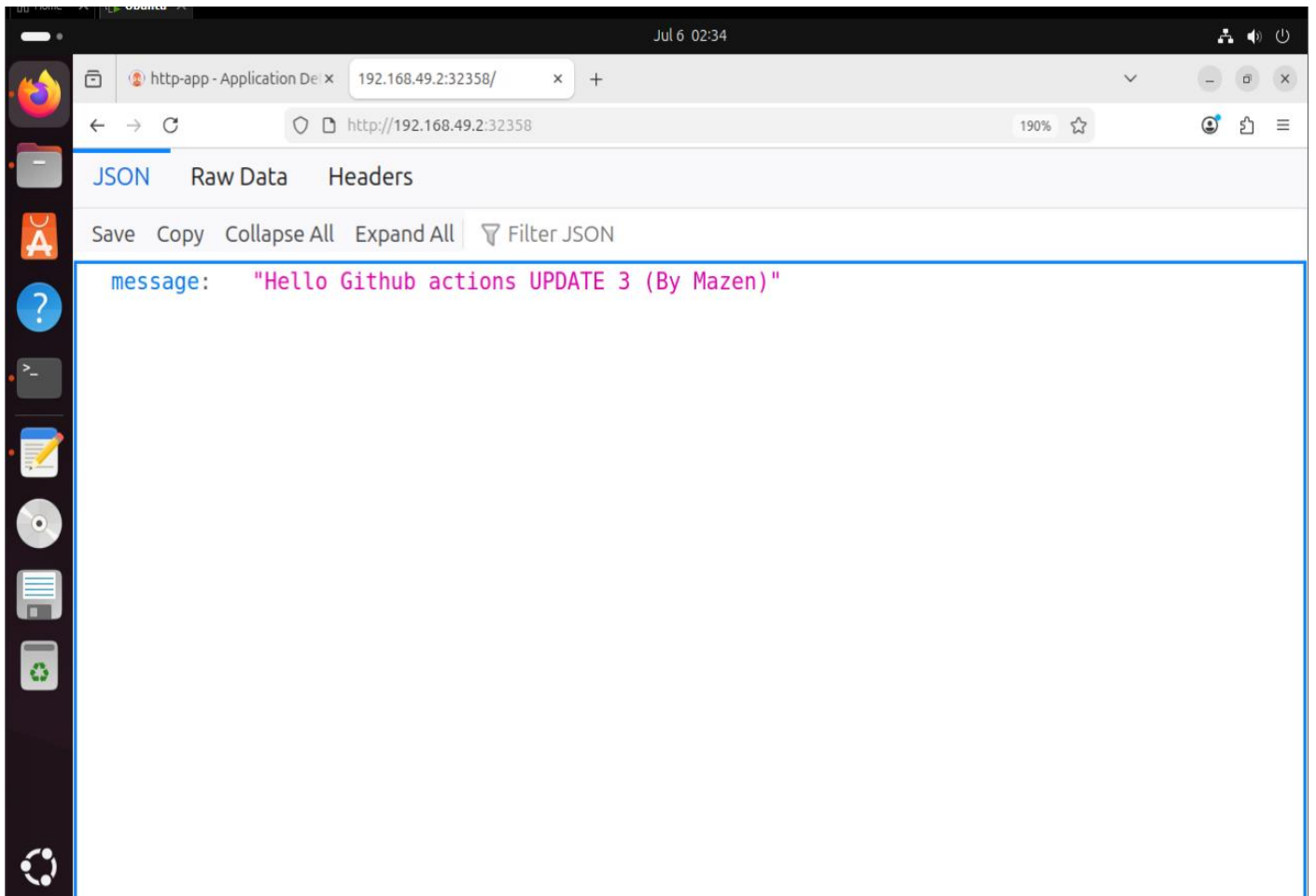
http-app

http-app-service

http-app-deployment

http-app-deployment-656db7...

http-app-deployment-dfb787...



MaZeNNaSSer22 / Ci-Using-GithubActions

Type to search

CodeIssuesPull requestsAgentsActionsProjectsWikiSecurity and qualityInsightsSettings

Ci-Using-GithubActionsPublic

PinWatch 0Fork 0Star 0

main1 Branch0 Tags

Go to file

Add file

Code

MaZeNNaSSer22 Change the http page content (check Cd functionality) 7a2066e · 4 minutes ago 14 Commits

.github/workflows	Change the image verion in manifest repo	35 minutes ago
src	Change the http page content (check Cd functionality)	4 minutes ago
.dockerignore	add (build & push to docker hub)	5 hours ago
.gitignore	first try	5 hours ago
Dockerfile	add (build & push to docker hub)	5 hours ago
README.md	first commit	6 hours ago
package-lockjson	first try	5 hours ago
package.json	first try	5 hours ago
tsconfig.json	first try	5 hours ago

About

This repository is used for practical exercises on Ci/Cd .

ReadmeActivity0 stars0 watching0 forks

Releases

No releases published
[Create a new release](#)

Packages

No packages published
[Publish your first package](#)

MaZeNNaSSer22 / Cd-Using-ArgoCD-Manifests-files-

Type to search

CodeIssuesPull requests1AgentsActionsProjectsWikiSecurity and qualityInsightsSettings

Cd-Using-ArgoCD-Manifests-files-Public

PinWatch 0Fork 0Star 0

main3 Branches0 Tags

Go to file

Add file

Code

MaZeNNaSSer22 Merge pull request #2 from MaZeNNaSSer22/update-image-13 b07066a · 1 minute ago 6 Commits

README.md	first commit	5 hours ago
deploy.yaml	chore: update http-app image tag to build 13	2 minutes ago
service.yaml	first try	4 hours ago

About

This repository is used for practical exercises on Ci/Cd .

ReadmeActivity0 stars0 watching

hub

ExploreMy Hub

Search Docker Hub

CtrlK

?

?

?

?

M

mazennasserYour personal account

RepositoriesHardened ImagesCollaborationsSettings

Default privacyNotificationsBillingUsagePullsStorage

Upgrade plan

Repositories / http-app / General

mazennasser/http-app

Last pushed 2 minutes ago · Repository size: 46.6 MB · 0 · 271

Add a descriptionAdd a category

Docker commands

To push a new tag to this repository:

docker push mazennasser/http-app:tagname

Public view

GeneralTagsImage ManagementCollaboratorsWebhooksSettings

Tags

DOCKER SCOUT INACTIVE

Activate

This repository contains 9 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	2 minutes
12		Image	less than 1 day	2 minutes
11		Image	less than 1 day	4 minutes
10		Image	less than 1 day	6 minutes
9		Image	less than 1 day	about 3 hours

buildcloud

Build with Docker Build Cloud

Accelerate image build times with access to cloud-based builders and shared cache.

Docker Build Cloud executes builds on optimally-dimensioned cloud infrastructure with dedicated per-organization isolation.

Get faster builds through shared caching across your team, native multi-platform support, and encrypted data transfer - all without managing infrastructure.

Go to Docker Build Cloud